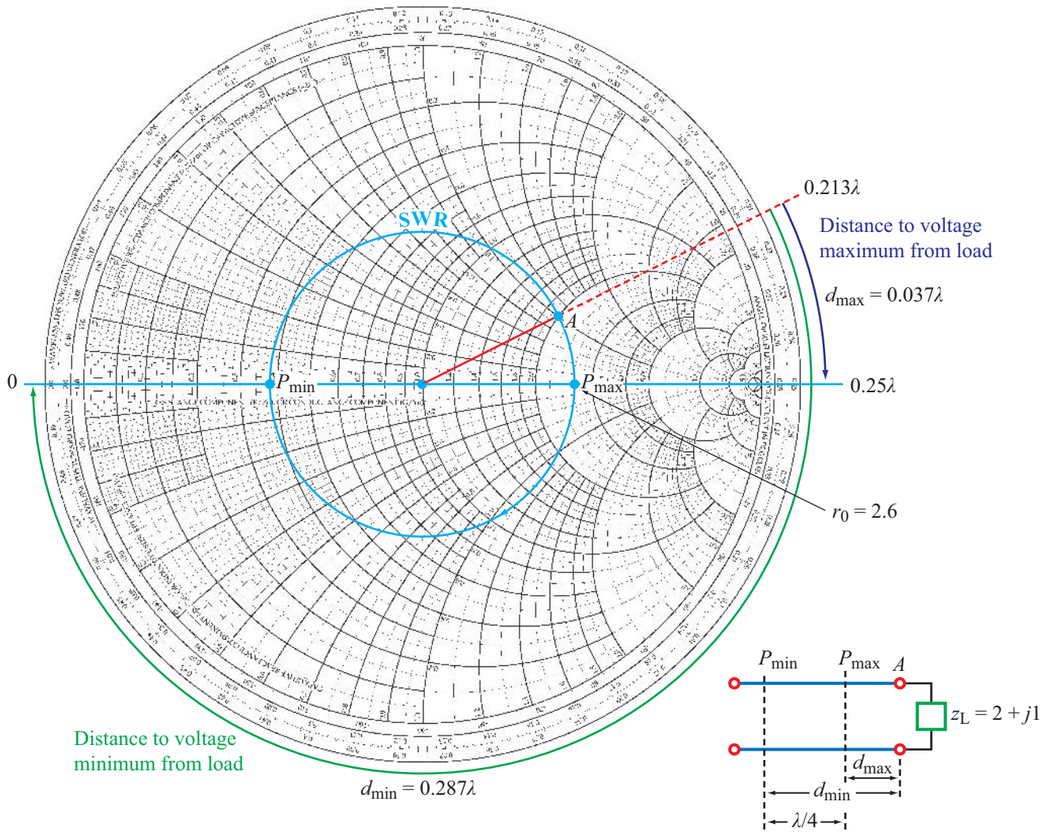


Figure 2-28 Point A represents a normalized load with $z_L = 2 + j1$. The standing-wave ratio is $S = 2.6$ (at P_{max}), the distance between the load and the first voltage maximum is $d_{max} = (0.25 - 0.213)\lambda = 0.037\lambda$, and the distance between the load and the first voltage minimum is $d_{min} = (0.037 + 0.25)\lambda = 0.287\lambda$.

Accordingly, using Eq. (2.122), the normalized load admittance y_L is given by

$$y_L = \frac{1}{z_L} = \frac{1-\Gamma}{1+\Gamma} \quad \textbf{(dimensionless)}. \tag{2.135}$$

Now let us consider the normalized wave impedance $z(d)$ at a distance $d = \lambda/4$ from the load. Using Eq. (2.120) with $2\beta d = 4\pi d/\lambda = 4\pi\lambda/4\lambda = \pi$ gives

$$z(d = \lambda/4) = \frac{1+\Gamma e^{-j\pi}}{1-\Gamma e^{-j\pi}} = \frac{1-\Gamma}{1+\Gamma} = y_L. \tag{2.136}$$

► Rotation by $\lambda/4$ on the SWR circle transforms z into y, and vice versa. ◄

In **Fig. 2-29**, the points representing z_L and y_L are diametrically opposite to each other on the SWR circle. In fact, such a

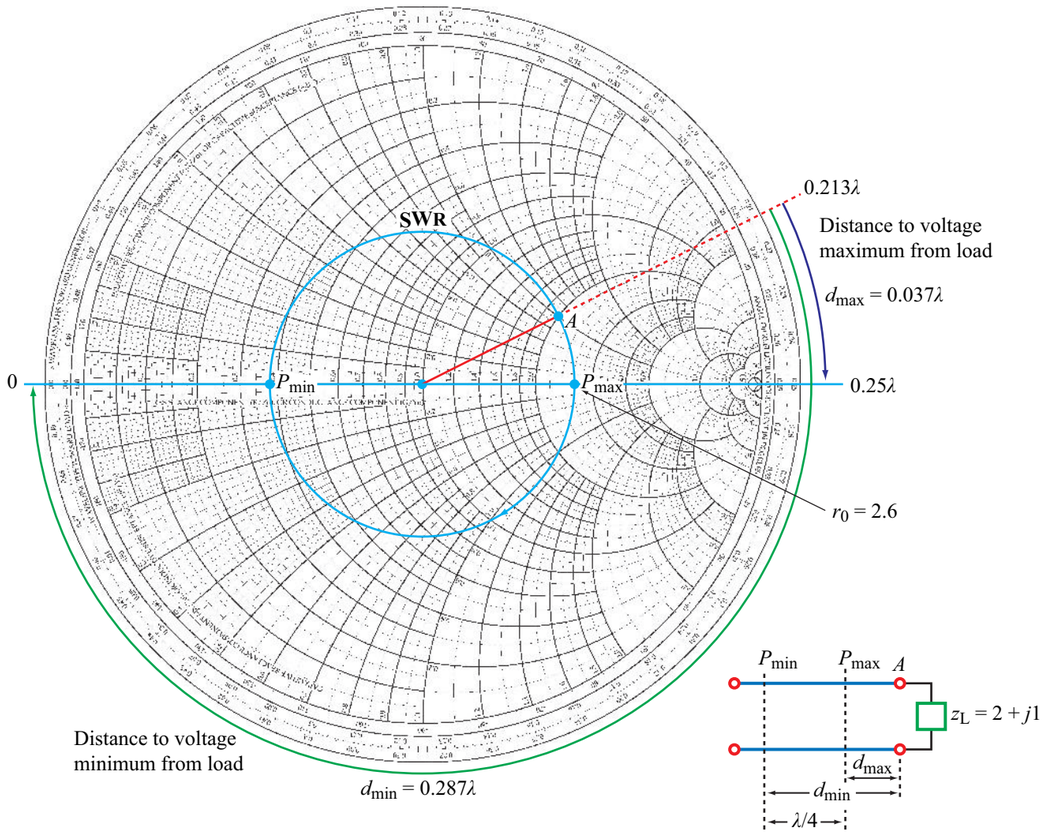


图 2-28 点 A 表示 $z_L = 2+ j1$ 的标准化载荷。驻波比为 S = 2.6（在 P_{max} 处），负载与第一电压最大值之间的距离为 $d_{max} = (0.25 - 0.213)\lambda = 0.037\lambda$，负载与第一电压最小值之间的距离为 $d_{min} = (0.037 + 0.25)\lambda = 0.287\lambda$。

因此，使用等式。(2.122)，归一化负载导纳 y_L 由下式给出

$$y_L = \frac{1}{z_L} = \frac{1-\Gamma}{1+\Gamma} \quad \textbf{(dimensionless)}. \tag{2.135}$$

现在让我们考虑距负载距离 $d = \lambda/4$ 处的归一化波阻抗 z (d)。使用方程式(2.120) 与 $2\beta d = 4\pi d/\lambda = 4\pi\lambda/4\lambda = \pi$ 给出

$$z(d = \lambda/4) = \frac{1+\Gamma e^{-j\pi}}{1-\Gamma e^{-j\pi}} = \frac{1-\Gamma}{1+\Gamma} = y_L. \tag{2.136}$$

► 在 SWR 圆上旋转 $\lambda/4$ 将 z 转换为 y，反之亦然。◄

在图2-29中，代表z_L和y_L的点在SWR圆上彼此相反。事实上，这样一个

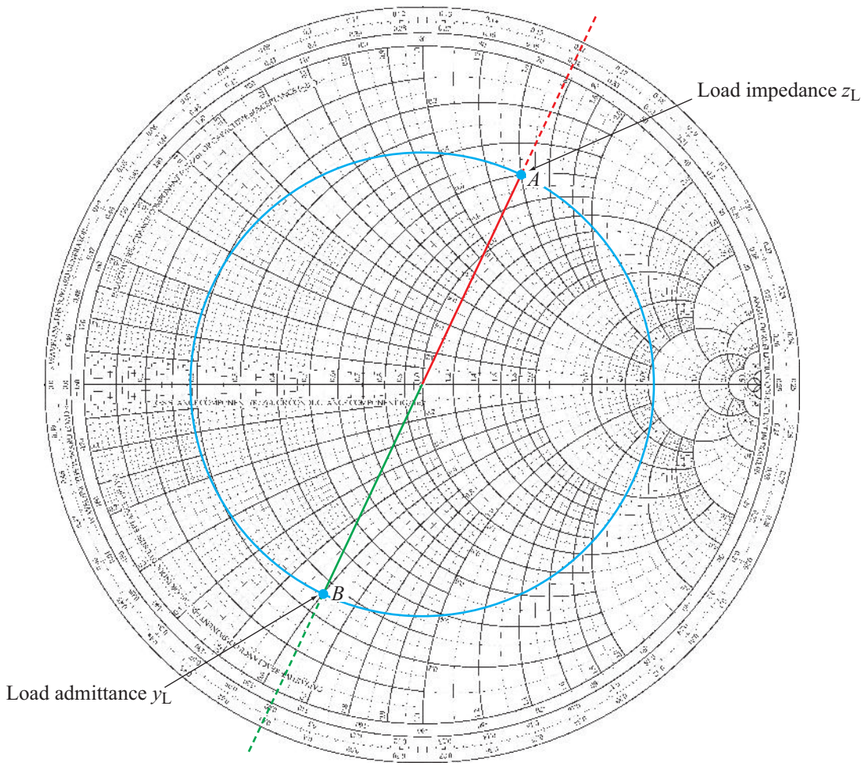


Figure 2-29 Point *A* represents a normalized load $z_L = 0.6 + j1.4$. Its corresponding normalized admittance is $y_L = 0.25 - j0.6$, and it is at point *B*.

transformation on the Smith chart can be used to determine any normalized admittance from its corresponding normalized impedance, and vice versa.

The Smith chart can be used with normalized impedances or with normalized admittances. As an impedance chart, the Smith chart consists of r_L and x_L circles: the resistance and reactance of a normalized load impedance z_L, respectively.

► When used as an admittance chart, the r_L circles become g_L circles and the x_L circles become b_L circles, where g_L and b_L are the conductance and susceptance of the normalized load admittance y_L, respectively. ◄

Example 2-11: Smith Chart Calculations

A 50 Ω lossless transmission line of length 3.3λ is terminated by a load impedance $Z_L = (25 + j50)$ Ω. Use the Smith chart to find (a) the voltage reflection coefficient, (b) the voltage standing-wave ratio, (c) the distances of the first voltage maximum and first voltage minimum from the load, (d) the input impedance of the line, and (e) the input admittance of the line.

Solution: (a) The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{25 + j50}{50} = 0.5 + j1,$$

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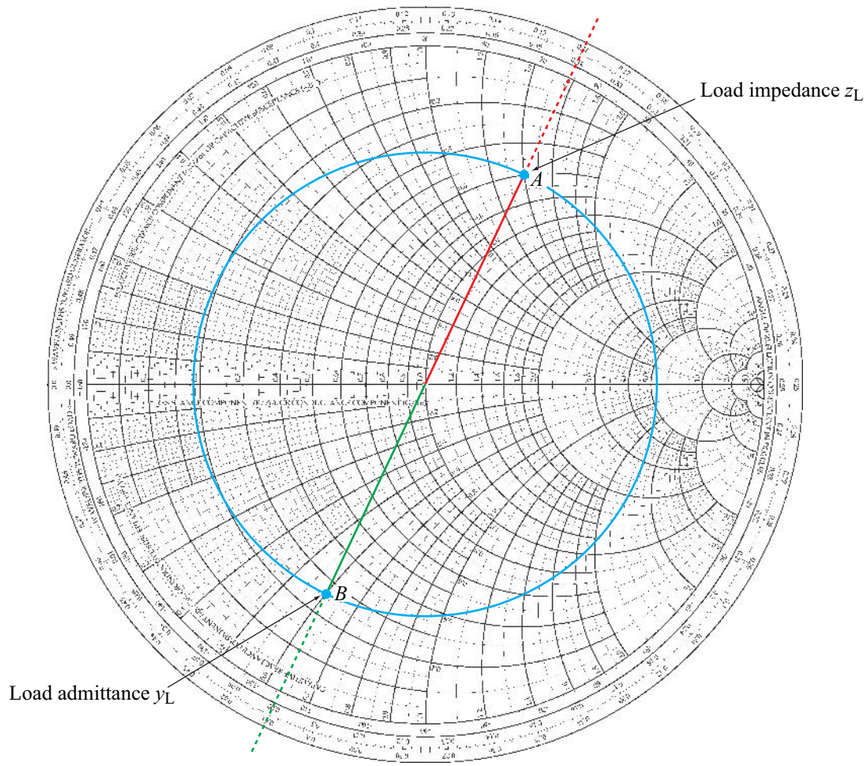


图2-29 A点表示归一化载荷$z_L = 0.6 + j1.4$。其对应的归一化导纳为 $y_L = 0.25 - j0.6$，位于 B 点。

史密斯圆图上的变换可用于根据其相应的归一化阻抗确定任何归一化导纳，反之亦然。

史密斯圆图可与归一化阻抗或归一化导纳一起使用。作为阻抗图，史密斯圆图由 r_L 和 x_L 圆组成：分别是归一化负载阻抗 z_L 的电阻和电抗。

► 当用作导纳图时，r_L 圆变为 g_L 圆，x_L 圆变为 b_L 圆，其中 g_L 和 b_L 分别是归一化负载导纳 y_L 的电导和电纳。 ◄

示例 2-11：史密斯圆图计算

长度为 3.3λ 的 50 Ω 无损传输线由负载阻抗 $Z_L = (25 + j50)$ Ω 终止。使用史密斯圆图求 (a) 电压反射系数，(b) 电压驻波比，(c) 第一电压最大值和第一电压最小值与负载的距离，(d) 线路的输入阻抗，以及 (e) 线路的输入导纳。

解： (a) 归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_0} = \frac{25 + j50}{50} = 0.5 + j1,$$

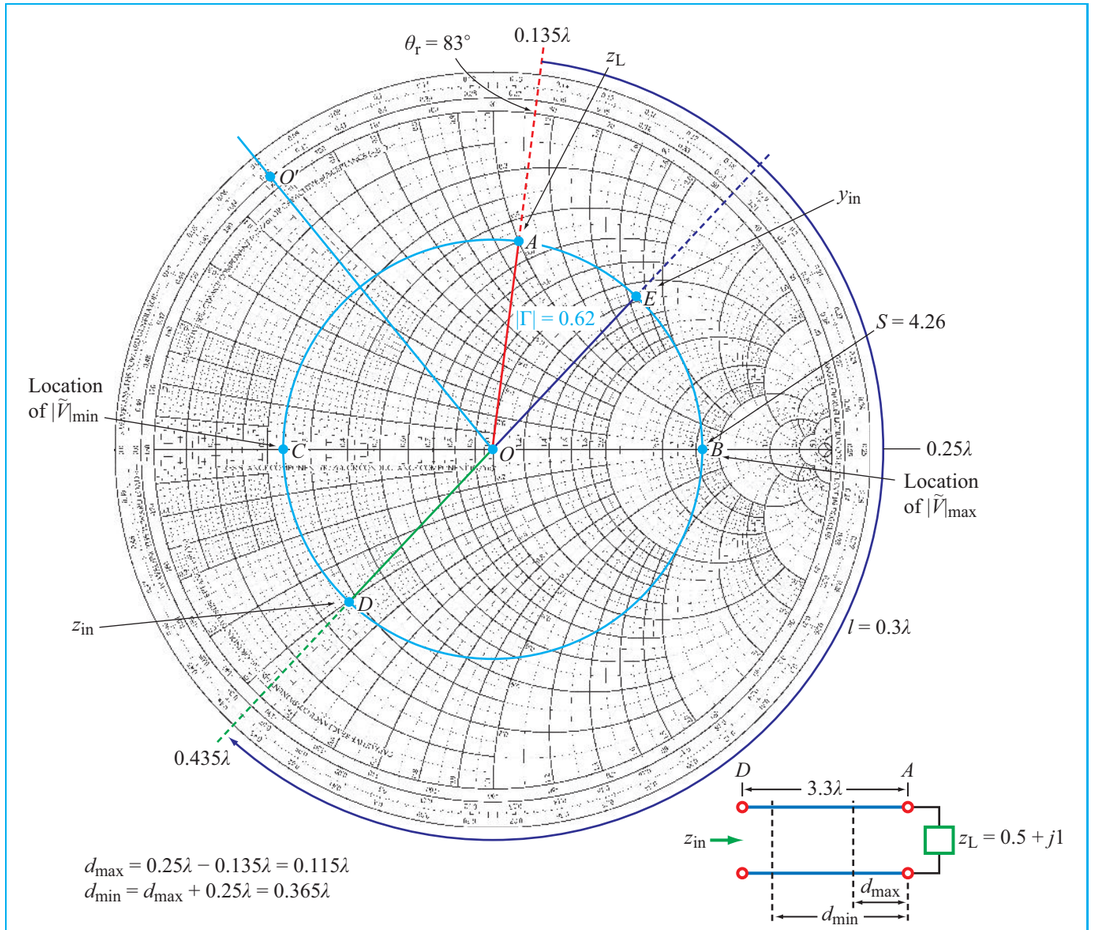


Figure 2-30 Solution for Example 2-11. Point *A* represents a normalized load $z_L = 0.5 + j1$ at 0.135λ on the WTG scale. At *A*, $\theta_r = 83°$ and $|\Gamma| = \overline{OA}/\overline{OO'} = 0.62$. At *B*, the standing-wave ratio is $S = 4.26$. The distance from *A* to *B* gives $d_{max} = 0.115\lambda$ and from *A* to *C* gives $d_{min} = 0.365\lambda$. Point *D* represents the normalized input impedance $z_{in} = 0.28 - j0.40$, and point *E* represents the normalized input admittance $y_{in} = 1.15 + j1.7$.

which is marked as point *A* on the Smith chart in **Fig. 2-30**. A radial line is drawn from the center of the chart at point *O* through point *A* to the outer perimeter of the chart. The line crosses the scale labeled "angle of reflection coefficient in degrees" at $\theta_r = 83°$. Next, measurements are made to determine lengths $\overline{OA}$ and $\overline{OO'}$, of the lines between *O* and *A* and between points *O* and *O'*, respectively, where *O'* is an arbitrary point on the $r_L = 0$ circle. The length $\overline{OO'}$ is equal to the radius of the $|\Gamma| = 1$ circle. The magnitude of Γ is then

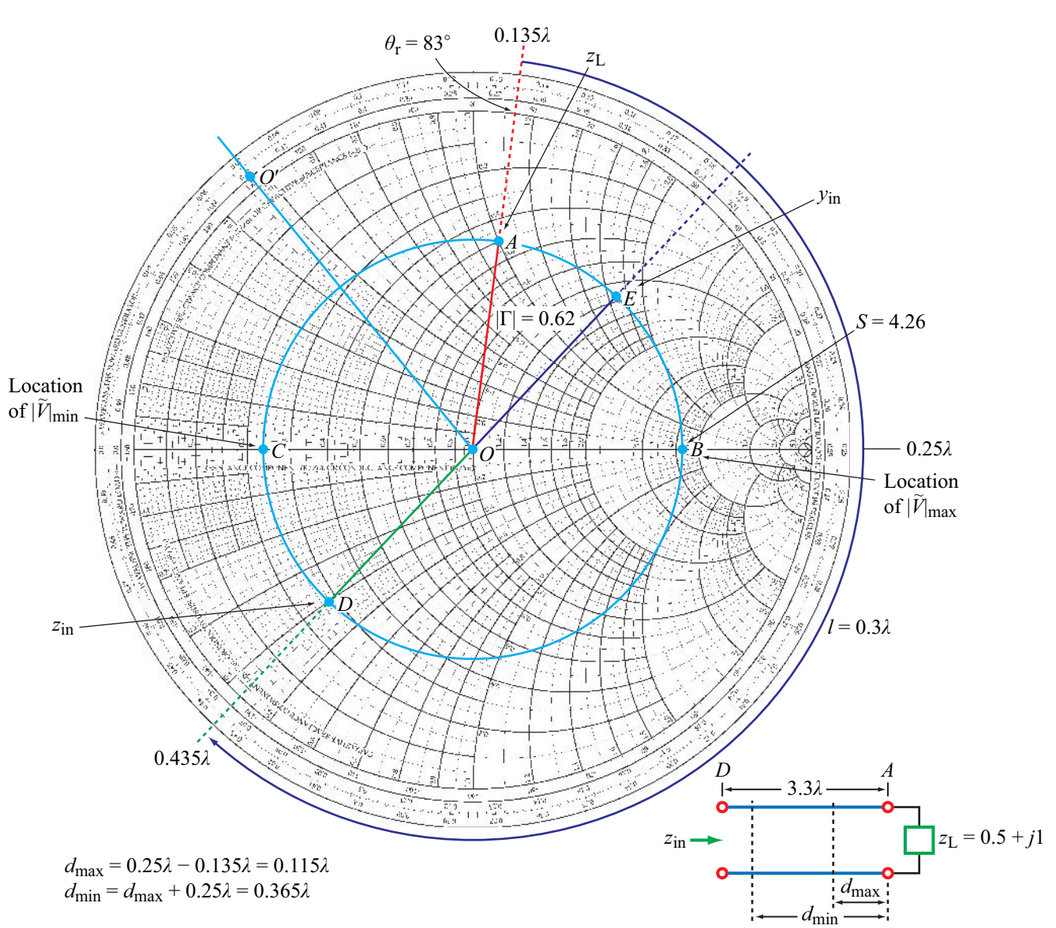


图 2-30 例 2-11 的解决方案。 A 点代表 WTG 规模上 0.135λ 处的归一化载荷 $z_L = 0.5 + j1$。在 A 处，$\theta_r = 83°$ 和 $|\Gamma| = OA/OO' = 0.62$。在 B 处，驻波比为 $S = 4.26$。从 A 到 B 的距离为 $d_{max} = 0.115\lambda$，从 A 到 C 的距离为 $d_{min} = 0.365\lambda$。 D点表示归一化输入阻抗z_{in} =0.28−j0.40，E点表示归一化输入导纳y_{in} =1.15+j1.7。

在图 2-30 的史密斯圆图上标记为 A 点。从图表中心的 O 点经过 A 点到图表的外周界绘制一条径向线。该线在 $\theta_r = 83°$ 处与标有“反射系数角度”的刻度相交。接下来，进行测量

分别确定 O 和 A 之间以及点 O 和 O′ 之间的直线的长度 OA 和 OO′，其中 O′ 是 $r_L = 0$ 圆上的任意点。长度OO′等于$|\Gamma| = 1$ 圆的半径。 Γ 的大小为

obtained from $|\Gamma| = \overline{OA}/\overline{OO'} = 0.62$. Hence,

$$\Gamma = 0.62\angle 83^\circ. \tag{2.137}$$

(b) The SWR circle passing through point A crosses the Γ_r axis at points B and C. The value of r_L at point B is 4.26, from which it follows that

$$S = 4.26.$$

(c) The first voltage maximum is at point B on the SWR circle, which is at location 0.25λ on the WTG scale. The load, represented by point A, is at 0.135λ on the WTG scale. Hence, the distance between the load and the first voltage maximum is

$$d_{\max} = (0.25 - 0.135)\lambda = 0.115\lambda.$$

The first voltage minimum is at point C. Moving on the WTG scale between points A and C gives

$$d_{\min} = (0.5 - 0.135)\lambda = 0.365\lambda,$$

which is 0.25λ past $d_{\max}$.

(d) The line is 3.3λ long; subtracting multiples of 0.5λ leaves 0.3λ. From the load at 0.135λ on the WTG scale, the input of the line is at $(0.135 + 0.3)\lambda = 0.435\lambda$. This is labeled as point D on the SWR circle, and the normalized impedance is

$$z_{in} = 0.28 - j0.40,$$

which yields

$$Z_{in} = z_{in}Z_0 = (0.28 - j0.40)50 = (14 - j20)\ \Omega.$$

(e) The normalized input admittance y_{in} is found by moving 0.25λ on the Smith chart to the image point of z_{in} across the circle, labeled point E on the SWR circle. The coordinates of point E give

$$y_{in} = 1.15 + j1.7,$$

and the corresponding input admittance is

$$Y_{in} = y_{in}Y_0 = \frac{y_{in}}{Z_0} = \frac{1.15 + j1.7}{50} = (0.023 + j0.034)\ \text{S}.$$

Example 2-12: Determining Z_L Using the Smith Chart

This problem is similar to Example 2-6, except that now we demonstrate its solution using the Smith chart.

Given that the voltage standing-wave ratio $S = 3$ on a 50 Ω line, that the first voltage minimum occurs at 5 cm from the load, and that the distance between successive minima is 20 cm, find the load impedance.

Solution: The distance between successive minima equals $\lambda/2$. Hence, $\lambda = 40$ cm. In wavelength units, the first voltage minimum is at

$$d_{\min} = \frac{5}{40} = 0.125\lambda.$$

Point A on the Smith chart in **Fig. 2-31** corresponds to $S = 3$. Using a compass, the constant S circle is drawn through point A. Point B corresponds to locations of voltage minima. Upon moving 0.125λ from point B toward the load on the WTL scale (counterclockwise), we arrive at point C, which represents the location of the load. The normalized load impedance at point C is

$$z_L = 0.6 - j0.8.$$

Multiplying by $Z_0 = 50\ \Omega$, we obtain

$$Z_L = 50(0.6 - j0.8) = (30 - j40)\ \Omega.$$

Concept Question 2-19: The outer perimeter of the Smith chart represents what value of $|\Gamma|$? Which point on the Smith chart represents a matched load?

Concept Question 2-20: What is an SWR circle? What quantities are constant for all points on an SWR circle?

Concept Question 2-21: What line length corresponds to one complete rotation around the Smith chart? Why?

Concept Question 2-22: Which points on the SWR circle correspond to locations of voltage maxima and minima on the line and why?

Concept Question 2-23: Given a normalized impedance z_L, how do you use the Smith chart to find the corresponding normalized admittance $y_L = 1/z_L$?

2-11 Impedance Matching

A transmission line usually connects a generator circuit at one end to a load at the other. The load may be an antenna, a computer terminal, or any circuit with an equivalent input impedance Z_L.

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从 $|\Gamma| = OA/OO' = 0.62$ 获得。因此，

$$\Gamma = 0.62\angle 83^\circ. \tag{2.137}$$

(b) 经过 A 点的 SWR 圆在 B 点和 C 点与 Γ_r 轴相交。 B 点的 r_L 值为 4.26，由此得出

$$S = 4.26.$$

(c) 第一个电压最大值位于 SWR 圆上的 B 点，即 WTG 标度上的位置 0.25λ。由 A 点表示的负载在 WTG 标度上为 0.135λ。因此，负载和第一电压最大值之间的距离是

$$d_{\max} = (0.25 - 0.135)\lambda = 0.115\lambda.$$

第一个电压最小值位于 C 点。在 A 点和 C 点之间移动 WTG 刻度可得出

$$d_{\min} = (0.5 - 0.135)\lambda = 0.365\lambda,$$

$d_{\max}$ 过去了 0.25λ。

(d) 线长 3.3λ；减去 0.5λ 的倍数得到 0.3λ。根据 WTG 标度上 0.135λ 处的负载，线的输入位于 $(0.135 + 0.3)\lambda = 0.435\lambda$。这被标记为 SWR 圆上的 D 点，归一化阻抗为

$$z_{in} = 0.28 - j0.40,$$

这产生

$$Z_{in} = z_{in}Z_0 = (0.28 - j0.40)50 = (14 - j20)\ \Omega.$$

(e) 归一化输入导纳 y_{in} 是通过将史密斯圆图上的 0.25λ 移动到穿过圆的 z_{in} 的图像点（SWR 圆上标记为点 E）来找到的。点E的坐标给出

$$y_{in} = 1.15 + j1.7,$$

相应的输入导纳为

$$Y_{in} = y_{in}Y_0 = \frac{y_{in}}{Z_0} = \frac{1.15 + j1.7}{50} = (0.023 + j0.034)\ \text{S}.$$

示例 2-12：使用史密斯圆图确定 Z_L

这个问题与示例 2-6 类似，只不过现在我们使用史密斯圆图演示其解决方案。

鉴于 50 Ω 线上的电压驻波比 S = 3，第一个电压最小值出现在距离

负载，并且连续最小值之间的距离为 20 cm，求负载阻抗。

解：连续最小值之间的距离等于λ/2。因此，λ = 40 厘米。以波长为单位，第一个电压最小值为

$$d_{\min} = \frac{5}{40} = 0.125\lambda.$$

图2-31中史密斯圆图上的A点对应于S= 3。使用圆规，通过A点画出常数S圆。B点对应于电压最小值的位置。在 WTL 标尺上从 B 点向负载移动 0.125λ 后（逆时针），我们到达 C 点，它代表负载的位置。 C 点的归一化负载阻抗为

$$z_L = 0.6 - j0.8.$$

乘以 $Z_0 = 50\ \Omega$，我们得到

$$Z_L = 50(0.6 - j0.8) = (30 - j40)\ \Omega.$$

概念问题 2-19：史密斯圆图的外周代表 $|\Gamma|$ 的值？史密斯圆图上的哪一点代表匹配负载？

概念问题 2-20：什么是 SWR 圆？对于 SWR 圆上的所有点来说，哪些量是恒定的？

概念问题 2-21：围绕史密斯圆图旋转一圈对应的线长是多少？为什么？

概念问题 2-22：SWR 圆上的哪些点对应于线上电压最大值和最小值的位置，为什么？

概念问题 2-23：给定一个归一化阻抗 z_L，如何使用史密斯圆图找到相应的归一化导纳 $y_L = 1/z_L$？

2-11 阻抗匹配

传输线通常将一端的发电机电路连接到另一端的负载。负载可以是天线、计算机终端或具有等效输入阻抗Z_L的任何电路。

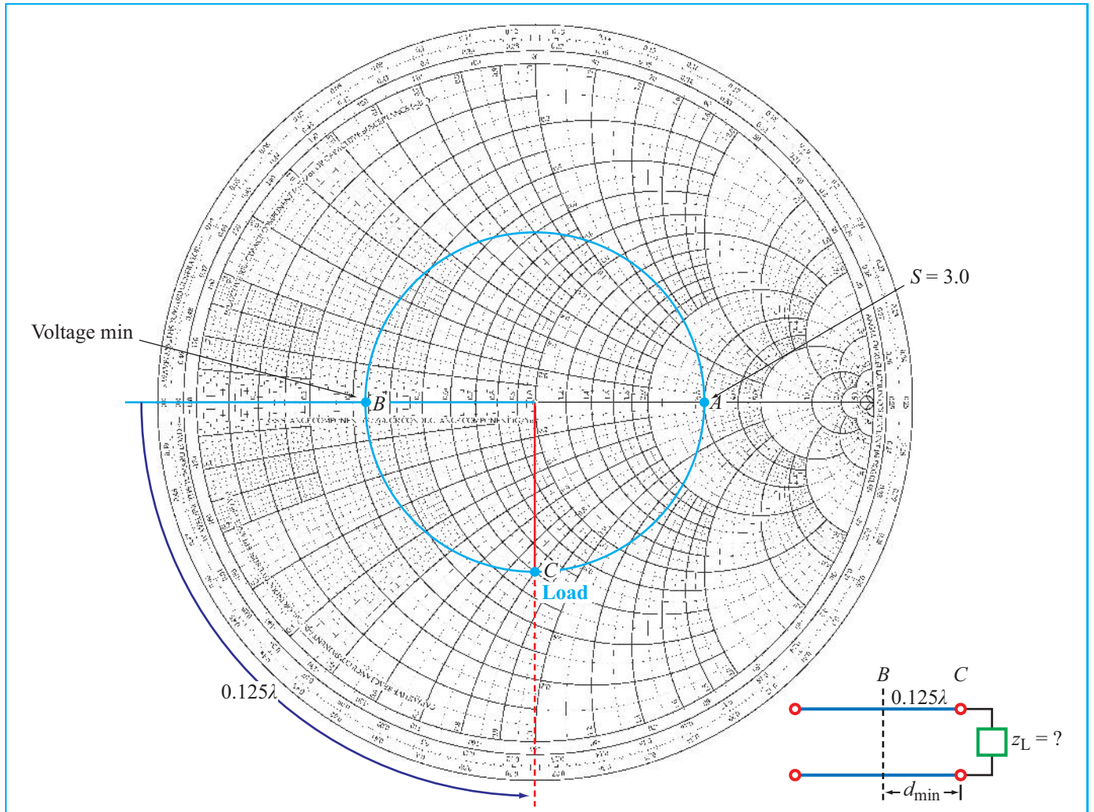


Figure 2-31 Solution for Example 2-12. Point *A* denotes that $S = 3$, point *B* represents the location of the voltage minimum, and point *C* represents the load at 0.125λ on the WTL scale from point *B*. At *C*, $z_L = 0.6 - j0.8$.

► The transmission line is said to be matched to the load when its characteristic impedance $Z_0 = Z_L$, in which case waves traveling on the line towards the load are not reflected back to the source. ◄

Since the primary use of a transmission line is to transfer power or transmit coded signals (such as digital data), a matched load ensures that all of the power delivered to the transmission line by the source is transferred to the load (and no *echoes* are relayed back to the source).

The simplest solution to matching a load to a transmission line is to design the load circuit such that its impedance $Z_L = Z_0$. Unfortunately, this may not be possible in practice because the load circuit may have to satisfy other requirements. An alternative solution is to place an ***impedance-matching network*** between the load and the transmission line as illustrated in **Fig. 2-32**.

► The purpose of the matching network is to eliminate reflections at terminals *MM′* for waves incident from the source. Even though multiple reflections may occur between *AA′* and *MM′*, only a forward traveling wave exists on the feedline. ◄

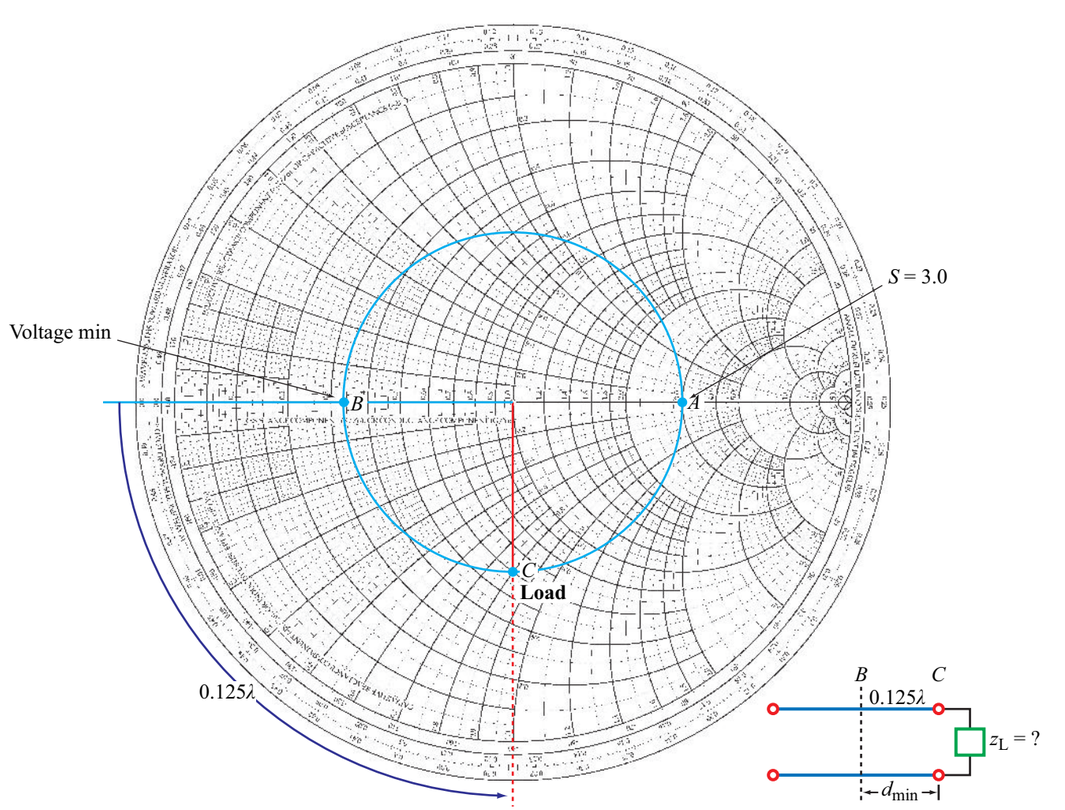


图 2-31 例 2-12 的解决方案。 A点表示S=3，B点表示电压最小值的位置，C点表示从B点开始的WTL标度上0.125λ处的负载。在C处，z_L =0.6−j0.8。

► 当传输线的特性阻抗 $Z_0 = Z_L$ 时，传输线被称为与负载匹配，在这种情况下，沿传输线向负载传播的波不会反射回源。 ◄

由于传输线的主要用途是传输功率或传输编码信号（例如数字数据），因此匹配的负载可确保源传递到传输线的所有功率都传输到负载（并且不会将回波中继回源）。

将负载与传输线匹配的最简单解决方案是设计负载电路，使其阻抗 ZL = Z_0。不幸的是，这在实践中可能是不可能的，因为负载电路可能必须满足其他要求。另一种解决方案是在负载和传输线之间放置一个阻抗匹配网络，如图 2-32 所示。

► 匹配网络的目的是消除终端 MM′ 对从源入射的波的反射。尽管AA′和MM′之间可能发生多次反射，但馈线上仅存在前向行波。 ◄